

TrES-3b

R-2799

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140713 · created 2026-09-04T23:40:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456851.8382 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.058
Transit depth (Rp/Rs)^2	1.83 +/- 1.56 [%]
Orbital Inclination (inc)	79.35 +/- 2.93
Airmass coefficient 1 (a1)	0.235 +/- 0.048
Airmass coefficient 2 (a2)	1.09 +/- 0.12
Scatter in the residuals of the lightcurve fit is	28.24 %
Best Comparison Star	#2 - [563, 159]
Optimal Aperture	1.85
Optimal Annulus	8.7
Transit Duration (day)	0.01 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.058 vs 0.16309 (deviation 0.4 σ)
Duration fit vs published	0.01 d vs 0.0567 d (deviation 2.93 σ)
Mid-time O-C	0.5 min
Depth SNR	1.9
Residual scatter	28.24 %

Light curve

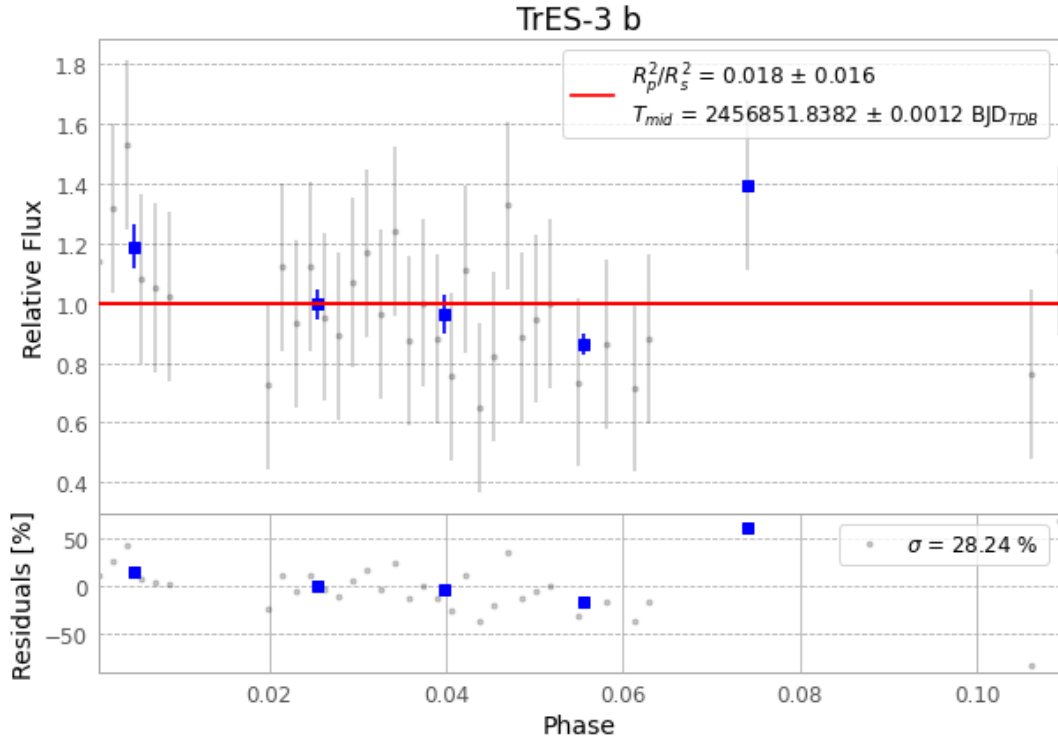


Figure 1 — FinalLightCurve_TrES-3 b_2014-07-13.png (SHA-256 ba0974eb3c3edffa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 271], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 94082dafcbc42a98...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), TRES-3140713080312.FITS → TRES-3140713112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140713.log (54f0135a45ba...), FinalLightCurve_TrES-3 b_2014-07-13.png (ba0974eb3c3e...), FinalLightCurve_TrES-3 b_2014-07-13.csv (a304f95f744e...)

Compute resources

Wall time 564.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 23:40Z.